

SIDDHANT DESHMUKH

Astrophysicist



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SUMMARY

Astrophysicist with a specialization in data science and machine learning. Over 5 years of experience working with statistical models, software development, computational modeling and data analytics. Effective communicator showcased through international conference presentations. Collaborative team leader with successful history of managing and delivering projects. Highly-motivated, team-oriented person with a problem-solving mindset.

KEY SKILLS

Programming Languages: Python, Rust, C, Julia, JavaScript, Java

Data Science: numpy, scipy, pandas, modeling, dimensionality reduction, forecasting, analysis

Machine Learning: TensorFlow/Keras, PyTorch, scikit-learn, classification, regression

Software: Linux, Git, GitHub, Microsoft Office, LaTeX

Soft Skills: Project management, presenting, report writing, supervising

Languages: English (fluent), German (working), Hindi (conversational), Marathi (conversational)

EXPERIENCE

10/2022 - 10/2023 **Max Planck Institute for Astronomy (sci-an)**

Lead Data Analyst

- Led project to create end-to-end text analysis pipeline for event data
- Implemented conventional data science and machine learning techniques
- Led social media marketing and targeted sales campaigns

08/2019 - 12/2022 **Heidelberg University**

Astrophysics PhD Researcher

- Led 2 independent research projects to publication
- Published thesis entitled "Modelling Non-Equilibrium Molecular Formation and Dissociation for the Spectroscopic Analysis of Cool Stellar Atmospheres"
- Incorporated machine learning techniques in tandem with physical modelling
- Secured funding for supervising 3-month student research project
- Presented research at multiple international conferences

05/2018 - 08/2018 **University of Exeter**

Undergraduate Researcher

- Created data processing pipeline for spacecraft observations of the solar wind
- Used Markov Chain Monte Carlo (MCMC) and regression models
- Developed preliminary forecasting method with LSTM neural networks

EDUCATION

2019 - 2023 **Heidelberg University**

PhD in Astrophysics

- Tutored modules in physics, computational astrophysics, and Python
- Worked in English & German

2015 - 2019 **University of Exeter**

Integrated Master's Degree in Physics (First Class with Honours)

- Computational modeling in Python & C for stellar magnetic field evolution
- Hosted student society outreach events and presented work